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Cosc 320-001

19 November 2019

## Lab 10 Read Me

### Summary:

This lab was not too long. I looked how to use vectors with iterators and the `std::map` functions, since I have yet to really use those. In addition, I followed the pseudo code for the Breadth First Search print function.

### Time Complexity:

The time complexity of BFS is  $O(|V| + |E|)$  where  $|V|$  is the number of vertices in the graph, and  $|E|$  is the number of edges in the graph. Depending on the size of the graph, these two variables differ. For example, the best case scenario is:

$$|E| = O(1) \rightarrow O(1)$$

But the worst case scenario is:

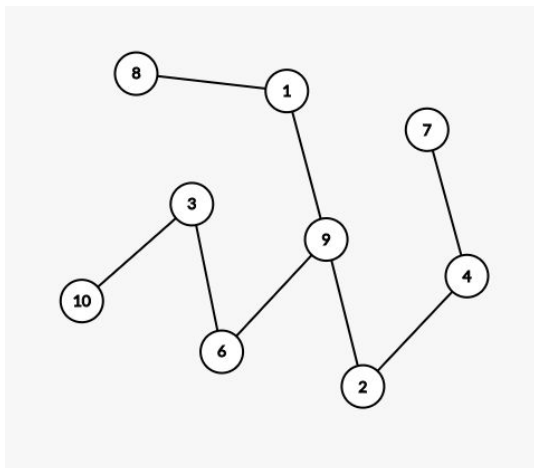
$$|E| = O(|V|^2) \rightarrow O(n^2)$$

### Code Improvement:

The code could have a small menu to demonstrate to the user the accuracy of the code (mainly the Breadth First Search).

### Program Graph Example:

Test 1



Test 2

